

KSHITIJ RAJ

BACKEND DEVELOPER | SYSTEMS | WEB SECURITY ENGINEER

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PROFESSIONAL SUMMARY

Performance-driven Backend and Systems Engineer experienced in architecting resilient distributed services, deterministic financial engines, and low-level cryptographic systems. Proven track record deploying production APIs supporting 1,000+ active users, porting post-quantum lattice algorithms to resource-constrained microcontrollers (<512 KB SRAM) at DRDO SAG, and hardening production Linux server infrastructure.

WORK EXPERIENCE

Robotic Sir AI India Pvt. Ltd. - Backend Developer and Server Security Engineer

June 2026 - Present | India

- Engineer production REST APIs across **25+ endpoints** with **Django**, **Node.js**, and **PostgreSQL** supporting **1,000+ active users**.
- Harden Debian Linux VPS infrastructure using **Fail2ban** for dynamic SSH brute-force jailing, custom **iptables/UFW** rules, and reverse proxy via **Nginx**.
- Maintain continuous health monitoring and scheduled database backup pipelines, reducing operational downtime risks and deployment overhead by **80%**.
- Optimize relational database schemas and indexed foreign keys, reducing p95 API response times by **35%** across high-traffic endpoints.

Scientific Analysis Group (SAG), DRDO - Student Intern (Post-Quantum Cryptography)

Jan 2026 - April 2026 | Delhi, India

- Deployed reference **NIST CRYSTALS-Kyber (ML-KEM)** post-quantum lattice algorithms on resource-constrained **ESP32** microcontrollers.
- Eliminated dynamic heap allocations (`malloc/free`) by designing in-place polynomial scratchpad buffers in **SRAM (<512 KB)**, preventing stack collisions.
- Validated cryptographic arithmetic and key encapsulation cycles across **1,000+ test iterations** against official **NIST Known Answer Test (KAT)** vectors, achieving a **100% pass rate**.

Nari Vastaram - Freelance Full-Stack Developer

April 2025 - July 2025 | Remote

- Delivered a production e-commerce platform using the **MERN Stack** (React.js, Node.js, Express.js, MongoDB) managing **50+ SKUs** and 100% of launch traffic.
- Designed REST APIs for shopping cart workflows, inventory tracking, and payment gateway checkout, reducing checkout drop-off by **18%**.
- Configured production services on a Linux VPS behind an **Nginx** reverse proxy with environment secret management, achieving **99.9% uptime**.

FEATURED PROJECTS

Follope - Invoicing and Real-time Payment Engine | TypeScript, Node.js, Express, PostgreSQL, Prisma, Redis, UPI

[follope.com](#)

- Architected a server-authoritative financial calculation engine with **deterministic integer-paise arithmetic**, eliminating IEEE-754 floating-point errors across taxes and line items.
- Secured public invoice viewing endpoints against **IDOR vulnerabilities** by generating collision-resistant **CUID tokens**, preventing enumeration of private ledger records.
- Integrated dynamic **NPCI-compliant UPI QR payload generation** and **HMAC-SHA256 signature verification** to safeguard webhook callbacks against tampering.
- Built a background sweep daemon dispatching push notifications via the **Expo Push API** and transactional NodeMailer OTP receipts.

School Management System (ERP) | TypeScript, Node.js, Express.js, MySQL, JWT, RBAC

[GitHub](#)

- Constructed a role-based access control (RBAC) educational ERP supporting administrators, instructors, and **1,000+ student records**.
- Normalized relational database schemas in **MySQL** with systematized fee invoicing, attendance tracking, and grading state machines.

WasteEx - B2B Industrial Circular Economy Platform | TypeScript, Node.js, MongoDB, Socket.IO, Web3

[GitHub](#)

- Developed a full-stack B2B marketplace for industrial materials trading with real-time negotiation channels powered by **Socket.IO**.
- Implemented blockchain smart contract escrow mechanisms for trustless payment milestones and continuous Cloudinary media pipelines.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), C, C++, SQL, Bash, HTML5/CSS3

Backend and Frameworks: Node.js, Express.js, Django REST Framework, Prisma ORM, REST APIs, WebSockets (Socket.IO), Celery

Databases and Caching: PostgreSQL, MySQL, MongoDB, Redis (Sliding Window Rate Limiting)

Security and Systems: Post-Quantum Cryptography (CRYSTALS-Kyber), Anti-IDOR (CUID tokens), HMAC-SHA256 Signatures, JWT/Argon2, Linux Server Hardening, Fail2ban, iptables

DevOps and Infrastructure: Docker, Nginx Reverse Proxy, Git/GitHub, CI/CD Workflows, AWS S3, Cloudflare, Linux VPS Administration

EDUCATION

Shobhit Institute of Engineering and Technology - B.Tech in Computer Science and Engineering

Aug 2023 - June 2027 | Meerut, UP

Relevant Coursework: Data Structures and Algorithms, Cryptography and Network Security, Database Management Systems, Operating Systems, Computer Networks, Software Engineering.

RESEARCH, HONORS AND CERTIFICATIONS

• **Research Publication:** Author, "Integrating Vedic Mathematics in Post-Quantum Cryptography" - Presented at National Conference (IKVI 2025); optimized polynomial Ring-LWE multiplication cycle counts by **34%** on embedded hardware.

• **Hackathon Award: Winner (1st Place)**, Internal Smart India Hackathon 2025 - Built an on-device privacy-preserving computer vision filter.

• **Algorithmic Milestones:** **220+ problems solved** across LeetCode and CodeChef (97% C++ Skill Assessment, ~59.2% acceptance rate across Arrays, Trees, Graphs, and DP).

• **Certifications:** Google Cybersecurity Professional Certificate, NPTEL Practical Cybersecurity, Pregrad MERN Stack Engineering.